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Module 5 Lab 3 - Devesh Parekh

```
clc
clear
```

Question 1

```
imgOri = imread('Lena128.bmp'); % read in image to imgOri variable
hFig = figure(1); set(hFig, 'position', [100 100 400 350]) % set the position
and size of our figure

subplot(3,3,1); imshow(imgOri), ; title('An 8-bit grayscale image'); % display
the original image in subplot 1

for k= 1 : 8
    subplot(3,3,10-k);
    X= bitget(imgOri,k);
    imshow(X, 'DisplayRange', [0,1])
    title(['BitPlane'])
end
trueSize
```

An 8-bit grayscale image



BitPlane



BitPlane



BitPlane



BitPlane



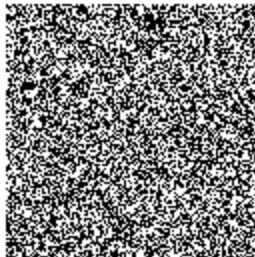
BitPlane



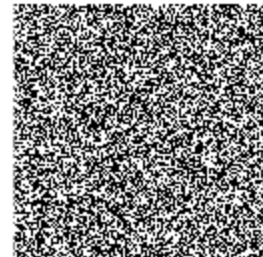
BitPlane



BitPlane



BitPlane



Question 2:

```
img1= imread('Lena512.bmp');  
hFig = figure(2); set(hFig, 'position', [100 100 400 350])  
subplot(2,2,1); imshow(img1), ; title('Original 512 bmp');  
for qualityFactor_vector = [100:-10:10];  
    filesize_vector = zeros(size(qualityFactor_vector));  
    imwrite(img1, 'Lena512.jpg', 'Quality' ,90 )  
    subplot(2,2,2);imshow('Lena512.jpg'), title('90% Quality')  
    imwrite(img1, 'Lena512.jpg', 'Quality' ,50 )  
    subplot(2,2,3);imshow('Lena512.jpg'), title('50% Quality')  
    imwrite(img1, 'Lena512.jpg', 'Quality' ,10 )  
    subplot(2,2,4);imshow('Lena512.jpg'),title('10% Quality')  
end
```

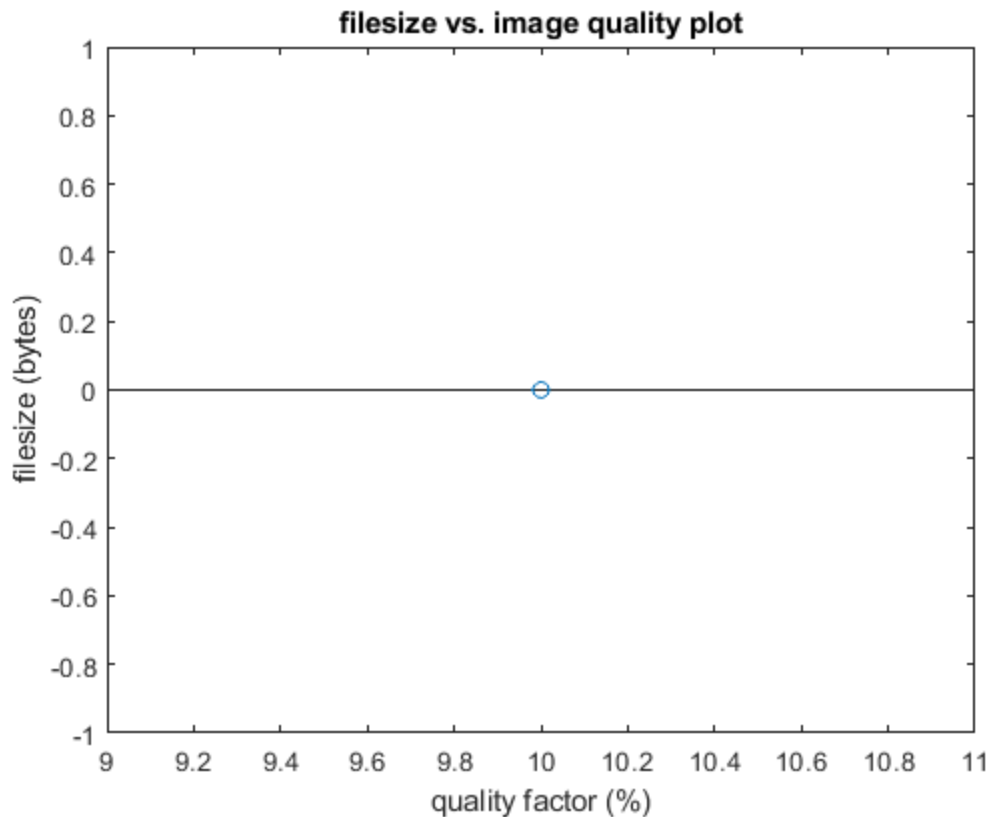


Question 3

The images seem to get worse in quality as the quality factor gets smaller and smaller

Question 4

```
figure(3);  
stem(qualityFactor_vector, filesize_vector);  
  
xlabel('quality factor (%)'); ylabel('filesize (bytes)'); title('filesize vs.  
image quality plot');
```



Question 5

for jpeg, The quality of the jpeg is determined by the file size

Question 6

% The image can be hidden in bit planes 0 through 4

Question 7

% for 90%, bitplane 3 is the most optimal

% for 50%, bitplane 7

% for 10%, bitplane 4

Question 8

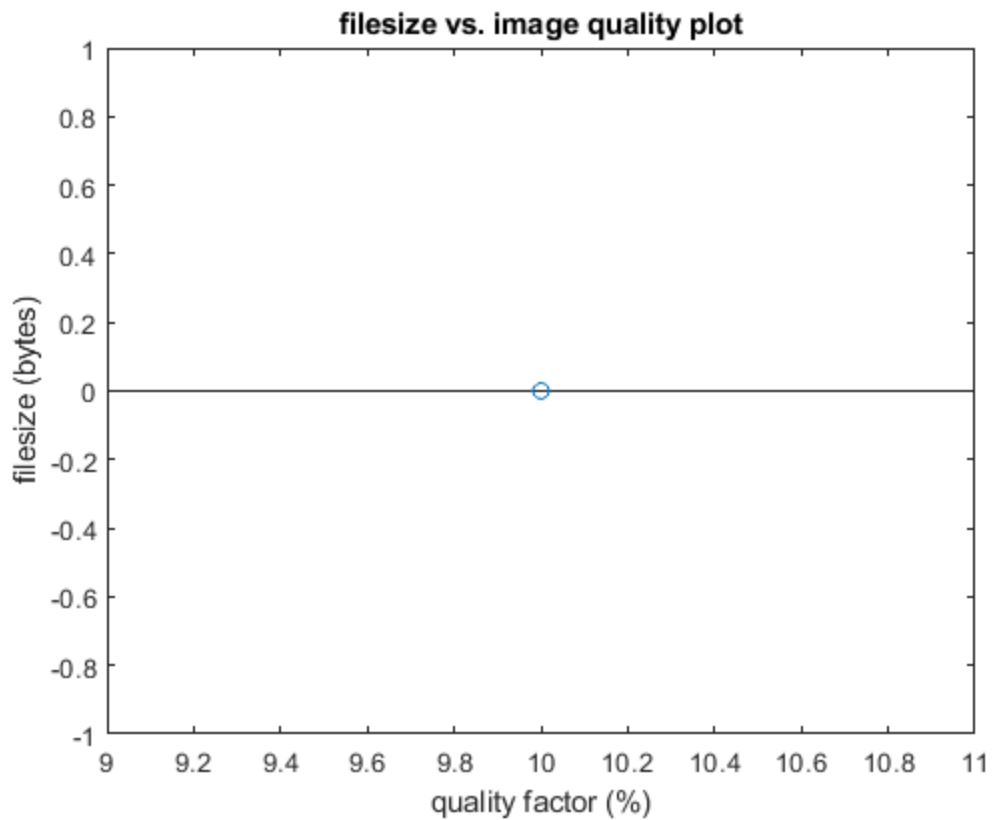
no i dont think so. Using a different image wont make a difference since you can just choose the same bitplanes to hide the images.

Question 12 from part 5

```
img_boat = imread('boat.tiff');
```

```
img_host = img_boat;
img_umdMark = rgb2gray(imread('UMD_mark_Athletic.tiff'));
img_toBeEmbedded = imresize(img_umdMark, 1/2, 'bicubic'); % reduce image size
by 1/2 in both height and width

figure(109); imshow(img_host); title('host image');
figure(110); imshow(img_toBeEmbedded); title('image to be embedded');
```



host image



image to be embedded

